

Microeconomic Analysis: Problem Set 4

Principal-Agent problems

Submit questions 1, 2 and 3

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1. Moral Hazard

A risk-neutral principal hires a risk-averse agent to undertake a project. There are two possible effort levels that the agent can exert, $e = 0$ or 1 , and three possible outcomes for the gross revenue to the principal, $\pi_1 < \pi_2 < \pi_3$. If $e = 0$, the probabilities of π_1, π_2, π_3 are $1/3, 1/3, 1/3$ respectively; if $e = 1$, the probabilities of π_1, π_2, π_3 are $1/4, 1/2, 1/4$ respectively.

The agent's utility from receiving a wage w and exerting effort e is $\sqrt{w} - e$, and the agent has a reservation utility of 10 .

In the contract between the two parties, w cannot depend on e directly, but can depend on the outcome, π_k . Thus a contract will be of the form $\{w_1, w_2, w_3\}$, where w_k is the wage paid by the principal to the agent if the gross revenue is π_k .

Assume that the gross revenues are such that the principal wants to induce the agent to choose $e = 1$.

- (a) Establish that $w_1^* = w_3^*$ in the optimal contract.
(You may assume that the K-T FOCs are necessary and sufficient for optimality. Also, there is no need to reproduce the general proof, but you should establish the result for this particular case.)
Hence, or otherwise, provide a complete characterisation of the optimal contract.
- (b) Suppose now that the agent can secretly destroy revenue.
 - i. Under the contract in part 1a, when might this be in the agent's interests?
 - ii. What is the new optimal contract, assuming that the principal does not want the agent to destroy revenue.
(An informal argument regarding an initial characterisation of wages is acceptable.)
- (c) Briefly, compare the agency costs from the contracts in parts 1a and 1b, and comment.

2. Monopolistic Screening

Blue-skies Travel monopolises air travel between Cloudy Bay and Pea-soup Island, transporting passengers at a cost of c each. There are two types of passenger, businessmen and tourists, a fraction λ being tourists, but the airline cannot directly tell them apart. Other things being equal, passengers prefer to buy their ticket as late as possible. Specifically, the utility each type gets from buying a ticket at a price P and a time W before the flight is:

$$v - \theta_b P - W \quad \text{or} \quad v - \theta_t P - W$$

where $0 < \theta_b < \theta_t$. (Note that for any given W , businessmen are prepared to pay more; also, businessmen are prepared to pay more for any given reduction in W .)

Assume in parts (a) to (d) that the airline wants to carry both types.

(For these parts, well-annotated diagrams could form the basis of your answer.)

- (a) In (P, W) -space, draw the indifference curves of the two types of passenger, and the isoprofit curves of the airline. Formulate the optimal price-discrimination problem that the airline wants to solve.
(Note that prices need to be constrained to be non-negative.)
- (b) Show that in the optimal solution, tourists are indifferent between travelling and not travelling.
- (c) Show that in the optimal solution, businessmen never buy their ticket in advance, and are indifferent between doing this and buying when tourists buy.
- (d) Describe fully the optimal price-discrimination scheme. How does it depend on θ_b/θ_t and λ ?
[Hint: There is a cut-off value for θ_b/θ_t in terms of λ on one side of which the airline prefers pooling to separating and *vice versa*.]
- (e) What should the airline do if c is higher than v/θ_b ? What should it do if c is between v/θ_t and v/θ_b ? For c below v/θ_t we might, in principle, see all passengers being served. However, even when costs are so low, find a condition in terms of c , v , θ_b , θ_t and λ under which the airline will not serve tourists.

3. Insurance – competitive, social planner, monopolist

- (a) Consider the Rothschild-Stiglitz model of competitive provision of insurance, where consumers are privately informed of their type, and the fraction of low-risk consumers is λ .

Explain why, in the candidate separating equilibrium, the high-risk consumers are offered full insurance.

Qualitatively, what is the condition on λ that ensures that equilibrium exists, and why?

- (b) Now assume that λ is such that a social planner can effect a Pareto improvement on the separating equilibrium above (without incurring a deficit).

What is the trade-off she considers?

In the Pareto optimal planner's solution, why are the high-risk consumers again offered full insurance?

Qualitatively, what is the condition on λ that ensures that a Pareto improvement is possible, and why?

- (c) Finally, consider the case where a risk-neutral monopolist is the insurance provider.

Explain what policies he will offer when λ is low (many high-risk consumers), and what policies he will offer when λ is high (many low-risk consumers).

Will the high-risk consumers yet again be offered full insurance?